Zihan Gong (Mr.)

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RESEARCH INTERESTS

Battery State-of-Health Estimation; Domain Adaptation and Generalization; Data-Efficient Learning; Dataset/Prototype Distillation; Generalized Foundation Models; Edge Intelligence for Battery Management Systems

EDUCATION

Xi'an Jiaotong University - Xi'an, China (C9 & Project 985&211) 09/2023-Present

Major: Intelligent Manufacturing Engineering (Qian Xuesen Honor Class, Automation Track) | GPA: 88.65/100

GitHub: https://github.com/asdstrind?tab=repositories Hugging Face: https://huggingface.co/coinlearner

PUBLICATIONS

First Author. "Task-Guided Prototype Distillation for Regression with Structure Preservation in Lithium-Ion Battery State-of-Health Estimation." IEEE Transactions on Emerging Topics in Computational Intelligence. (**Under review**)

Wending Zhu, Xiang Li, **Zihan Gong**, Mamdouh Abdelsalam, Wei Zhang, and Shaojie Yang. "A Large Multimodal Foundation Model for Unified State-of-Health and Remaining Useful Life Estimation of Lithium-ion Batteries." Journal of Energy Storage 177 (2026): 123418.

Zihan Gong, Xiang Li, Wei Zhang, and Shaojie Yang. "Generalized Foundation Model for Lithium-ion Battery State-of-Health Prediction with Distribution Metric Learning." Journal of Energy Storage 150 (2026): 120566.

RESEARCH EXPERIENCES

First Author, Task-Guided Prototype Distillation for Regression with Structure Preservation in Lithium-Ion Battery State-of-Health Estimation *(Under Review)*, *Advisor: Xiang Li* 02/2026-Present

Background

- Large-scale battery aging data contain myriad similar charge-discharge trajectories; using the full dataset increased storage, training time, and computational cost. Traditional dataset distillation methods were primarily designed for 2D image classification and could not directly handle 1D time series with continuous SOH labels. The objective was to compress large-scale real cycles into a small number of learnable data prototypes while preserving regression relationships, cross-sample distribution, and degradation sequence structure.

Methodology and Implementation

- Concretized "data reduction" into three simultaneous objectives, namely preserving SOH label semantics, overall distribution, and temporal structure, and organized loss functions, ablation studies, and visualization analyses around these objectives.
- Took charge of formulating research questions, designing the methodology, implementing the prototype generation pipeline, organizing experimental comparisons, and completing the main manuscript. Constructed a coherent narrative that wove method motivation, algorithm pipeline, experimental setup, and limitations, clarifying the correspondence between each constraint and its targeted failure mode.
- Treated data prototypes as learnable variables as opposed to selected real samples. First adopted SOH anchors to assign health state semantics to each prototype and used Huber regression loss to constrain the predictive relationship. Then employed Maximum Mean Discrepancy (MMD) to align synthetic prototypes with the original data distribution in feature space. Finally incorporated similarity constraints and 2D total variation regularization to suppress signal drift and local high-frequency artifacts.
- Implemented the pipeline by transforming voltage, current, and time sequences into Gramian Angular Field (GAF) images, extracting features via a fixed ResNet-2D evaluation network, and updating prototypes through joint loss back-propagation. Conducted experiments on HNEL and Toyota-MIT datasets against Random, Stratified, and Full-data baselines.

Results

- In the HNEL experimental setup of the submitted manuscript, an MAE of 0.0192 was achieved when using three synthetic prototypes, outperforming the corresponding Random and Stratified sampling baselines, and maintaining degradation trends more stably than the same number of real samples under extreme compression conditions.

Python-based Data Cleaning, Downsampling, and Unified Storage Format for Multi-source Lithium-ion Battery Data, ***Advisor: Xiang Li*** 03/2026-04/2026

Journal of Energy Storage 177 (2026): 123418.

Background

- Multi-source heterogeneous lithium-ion battery data exhibited significant discrepancies in cycle length, field format, sampling frequency, and file organisation. A unified and reproducible data interface needed to be established to support cross-domain training of multimodal large models.

Methodology and Implementation

- Used Python to read battery data from various raw storage formats (e.g., .csv, .mat) and removed outliers.
- Down-sampled all raw sequences uniformly to 256 time points, consistent with the data preprocessing pipeline in the paper. Eliminated discrepancies in length and sampling rate across different data sources. (Normalization was optionally applied as needed.)
- Serialized the cleaned data into .pkl format. Organized the directory structure by battery ID and cycle number. Stored each data domain as a separate dictionary.
- Generated battery data with a unified format across all datasets, with a data structure that enabled future extensibility.
- Wrote a script to verify sample counts, dimensions, and read accuracy item by item, ensuring the completeness and consistency of the delivered data.

Results

- Delivered a standardized training dataset that supported the team's Qwen 3.0B multimodal large model in performing LoRA-based parameter-efficient fine-tuning and unified SOH/RUL estimation across 11 heterogeneous datasets.
- Based on this data, the overall model achieved an SOH MAE of 0.0195 and an RUL MAE of 0.0961 on unseen batteries, and was able to identify early signs of battery degradation via anomaly detection. Data preprocessing removed the interference of original discrepancies on model training, making the training pipeline plug-and-play and providing a reliable infrastructure for cross-domain generalization and subsequent replication.

Generalized Foundation Model for Lithium-ion Battery State-of-Health Prediction with Distribution Metric Learning, ***Advisor: Xiang Li*** 09/2024-12/2025

Journal of Energy Storage 150 (2026): 120566.

Background

- Battery aging datasets for new energy vehicles exhibited significant distribution discrepancies across different material systems, cycling protocols, and operating conditions. The target domain typically contained only a limited number of labeled batteries, causing models to be biased toward the average degradation trend and fail to align with the target battery. This research aimed to learn transferable degradation representations with few-shot target-domain labels while simultaneously respecting the real-time computational constraints of battery management systems (BMS).

Methodology and Implementation

- Collected 11 publicly available battery aging datasets (covering different material systems and operating conditions, including HNEL, TongJi, Toyota-MIT, etc.), retaining cycles with SOH > 80%, downsampled voltage, current, and time sequences to length 256 per cycle, whilst preserving raw amplitude information (without uniform normalization).
- Innovatively designed a lightweight CNN-Transformer-KAN hybrid architecture: CNN for local temporal feature extraction, Transformer for long-range dependency modeling via self-attention, and KAN for flexible nonlinear regression.
- Jointly optimized SOH regression error and Maximum Mean Discrepancy (MMD) constraint to align feature distributions across different domains in the feature space.
- Conducted experiments with source-target splits on three different datasets for cross-domain validation, comparing against Target-only, Single-source, CNN, and CNN-Transformer baselines; also evaluated parameter count, computational cost (FLOPs), and single inference time to assess BMS deployment feasibility.

- Independently completed the entire pipeline from problem formulation, data curation, preprocessing, model implementation, training with alignment, comparative experiments, result analysis, to paper writing.

Results

- Reduced cross-domain prediction error from Target-only baseline (MAE > 3.8%, RMSE > 4.0%) to MAE < 1.1%, consistently within 1% magnitude across different target datasets.
- Achieved lightweight and real-time performance with 8.37M parameters, 23.11 MMac FLOPs, and 8.9 ms inference time, meeting BMS real-time constraints.
- Verified feature distribution alignment via t-SNE visualization, and confirmed the contribution of each module through ablation studies.
- Demonstrated the feasibility of balancing cross-domain accuracy with BMS real-time performance.

COMPETITIONS

2025 China Undergraduate Mathematical Contest in Modeling (CUMCM) *Problem B: Optimization of Smoke Jamming Projectile Launch Strategy Based on a Coupled Multi-Algorithm Framework (**Provincial First Prize**)* 09/2025

- Participated in constructing a kinematic model for missile-UAV-smoke projectile systems; employed a line-segment-to-sphere intersection criterion to determine line-of-sight occlusion status; for high-dimensional non-convex optimization problems, employed a PSO-GA coupling strategy to balance global and local search.
- Implemented exhaustive search, PSO, GA, and the coupled optimizer in C++ with the standard library (iostream, cmath, algorithm), and conducted algorithm comparison and parameter analysis.
- Generated convergence curves and result charts, translated computational logic and numerical results into figures and textual descriptions for the paper, and participated in manuscript writing.

2025 Mathematical Contest In Modeling/Interdisciplinary Contest In Modeling (MCM/ICM) *Problem E: Green Agriculture: Future or Not? (**Meritorious Winner**)* 01/2025

- Contributed to the development of a multi-species reaction-diffusion model incorporating Lotka-Volterra-type predation, competition, and cooperation; assisted with sensitivity analysis and variable consistency checks.
- Performed numerical simulations, parameter sweeps, and visualization in MATLAB R2023a; co-authored the competition manuscript and typeset in LaTeX.

INTERNSHIP

Intern, IT R&D Department of Geely Automobile Xi'an Manufacturing Base, Xi'an, China 07/2026

- Participated in the implementation and validation of a single-template retrieval-based wheel hub recognition scheme; gained familiarity with the experimental logic and data processing pipeline of WheelPrint and RSM-HOG feature fusion and the rejection mechanism; reviewed the Python code implementations (NumPy, scikit-image, pandas, etc.) for preprocessing, descriptor extraction, recognition, and evaluation modules.
- Authored the project report and technical documentation, synthesizing algorithm workflow, experimental design, and test results into a coherent narrative and forming a closed-loop documentation chain covering "design-test-diagnosis-improvement.", and participated in validating experimental data and summarizing misclassification cases caused by strong light reflections.
- Achieved a Top-1 accuracy of 97.62% on the development set with the fusion scheme; with rejection enabled, obtained 100% accuracy on accepted samples (79.76% coverage) and a median latency of 226 ms per image on CPU. The results were compiled into a technical report to support subsequent deployment evaluation.

SKILLS

- Rich experience in data-efficient and generalizable deep learning for battery health management and engineering time-series analysis.
- Familiar with domain adaptation, dataset/prototype distillation, multimodal foundation models, transfer learning, and scientific machine learning using Python and PyTorch.
- Knowledge of MATLAB, edge AI deployment, embedded systems, signal processing, and battery state-of-health estimation.